Engineering Mechanics 1 · Unit_01 · UNIT 1

DRAFT — UE26ME142A · Unit_01 · UNIT 1 · UE26ME142A Unit 1 Class 5 AVSummary

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

UE26ME142A Unit 1 Class 5 AVSummary

1 UE26ME142A Unit 1 Class 5 AVSummary

Study focus: UE26ME142A_Unit 1_Class 5_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000126 · EV0000127

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

I am Dr.

Key details

- Venkitish from Civil Engineering under Department of Science and Humanities, PS University.
- This is class number eight.
- Now today we'll continue with that and we will discuss the next topic, which is on moment.
- What is a moment?
- The definition of a moment is nothing but.
- The rotating effect produced by a force on a body.
- When a force is applied on a body, if the body rotates, that rotation of the body is called as a moment.
- One is translatory motion, the other one is the rotatory motion.
- The translatory motion is the displacement.
- When a force is applied onto a body, the body would move from a point A to a point B, or when a force is applied, the body would rotate.
- Without any displacement, it will be at the same point, but it will rotate.
- The rotating effect caused by the force is termed as moment.
- Mathematically, we represent moment as M is equal to R into F.
- or F into R if we are solving it by scalar.
- But if we are solving it in vector approach, then it follows what is called as a cross product.
- In earlier classes I had also discussed about cross product.
- I have talked about why it should be the position vector cross force vector and not force vector cross position vector.

- The distributed property is not valid in case of cross product.
- Position vector cross the force vector.
- Now in scalar approach it can be F into R or R into F, it does not matter because there we have to take the rotation into account and then decide whether it is positive or negative.
- But when we are using cross product.
- By doing or executing the cross product, we would get the sign for the momentum.
- Ringe is an instrument which is used to rotate a pipe or a bolt or a net.
- Which is perpendicular to this tool, which is called as a range.
- The magnitude of the rotation is going to be equal to the force that is applied into the perpendicular distance between the point where we want the moment.
- And the point where the force is applied.
- That distance in this case is B.
- Has to be 90 degrees, then only this distance would be the perpendicular distance.
- If you orient your thumb, right hand thumb.

Example or application from the lecture

- In the last class we had solved a few problems on resolution of forces, splitting up of a given force along its ax along its x and y axis.
- And determining the resultant of the given force.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000126 · EV0000127 · EV0000128 · +1 source units in pack.json

Core explanation · 05:00–09:59

Along the positive axis on which or about which the movement is caused, and if the other four fingers, as you can see, is folded like this.

Key details

- If that rotates, if that rotates, then.
- Like this, okay, as indicated in this diagram, then that would be the positive sign for the moment also.
- Orient the thumb, orient the thumb along the positive direction of the axis about which the moment or the rotation is occurring.
- If the rotation occurs similar to the other.
- Fingers of the right hand as indicated here, then we take that as positive.
- That is this axis.
- I want to find out what is the moment.
- If this distance is R, this angle is alpha, either I can resolve this force.
- Which is perpendicular to the axis and parallel to the axis, only the component of the force which is perpendicular to this axis would produce a moment.
- F sin alpha into R would be the magnitude of M, that is one way of solving it.
- The other way of solving it is you extend this line and determine what would be the perpendicular distance of this force.
- From point A, which is indicated here as D.

- This is 90 degrees.
- If this is alpha, then this would also be equal to alpha.
- We know that this is R.
- This forms a right angle triangle.
- Whether you resolve the force or you determine the perpendicular distance of the force with the point about which it is rotating.
- Or about which we are finding out the moment, we get the same value.
- Force is F and the perpendicular distance is D.
- The moment produced is the force into this perpendicular distance D.
- If this is x-axis and this is y-axis, the body is in xy plane, then the movement would be produced in an axis which is perpendicular to xy.
- The thumb of my right hand towards me, then you can see that the other four fingers are rotation.
- That rotation will be the positive direction of the moment, and that is how it is rotating.
- Waring Nord's theorem states that the moment produced by a force is equal to the moment produced by the force.
- Components about the same point.
- I'll repeat.
- Moment produced by a force, in this case, if I take R as the force.
- The movement produced by this R about any point, now I will take point O.

Example or application from the lecture

- If the rotation occurs opposite to this, that is like this, then we consider it to be negative.
- Say I will take the same example, an axis is there.
- Next, let's look at what is Warring Nord's theorem.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000130 · EV0000131 · EV0000132 · +1 source units in pack.json

Core explanation · 10:00–14:59

That should be equal to the component of this force R, which in this case is taken as P and Q about some axis.

Key details

- We cannot represent a force by about any other axis which will produce the same effect.
- No, at least there should be two axis about which we can find out the components.
- The moment produced by this R about O should be equal to the moment produced by Q about O plus the movement produced by P about O.
- It is the algebraic sum.
- Not just the sum, it is algebraic sum.
- B you have a movement produced by R above O will be R into B.
- That would be equal to your now.
- How what is the sense of this?
- The R is acting like this.

- The point O is here, so it makes a rotation.
- Like this, that is clockwise rotation.
- Clockwise rotation we take it as negative, and anti-clockwise rotation we take it as positive.
- Or vice versa, you can take it vice versa also.
- But usually when we go for vector approach, that is the cross product, we get clockwise as negative and anti-clockwise as positive.
- That should be equal to your P into the perpendicular distance of P.
- This is the perpendicular distance, which is considered as small Q.
- This force P and this perpendicular distance are small P would produce an anti-clockwise movement.
- That is what Varingov's theorem states.
- What is Varinghaus theorem states?
- Moment produced by a force about any point is equal to the moment produced by the force components about the same point.
- The statement of the problem is calculate the magnitude of the moment about the base O of the 600 newton force in five different ways.
- How do I get the moment of this force, 600 newtons, about this point O in five different ways.
- Then if you read the diagram.
- This diagram gives you the position of A.
- It is at a distance of 2 meters.
- If I take this as x-axis.
- And this has y-axis, okay.
- 2 meters positive x and the origin O is 4 meters from the intersection of these two lines.
- From O to A, the horizontal distance is 2 meters.
- Then the force is inclined at an angle of 40 degrees and the sense of the force is moving away from A, like this.
- This is the same line of action of the force.
- We have to take a moment about this.
- One way is take this force, determine what would be the perpendicular distance of this force from O.
- If I can find that then that will be equal to 600 into D, that is one way of solving it.
- The next way of solving it is resolve this force along this and along this.
- Take movement of this about O, take moment of this about O.
- Next, extend this line and see to it that it cuts this axis at some point.
- Resolve this force along horizontal.
- Determine what is the distance from here to here.

Example or application from the lecture

- We can resolve a given force along at least along two axis, not lesser than that.
- To have the consistency, let us follow the same thing, we will have anti-clockwise as positive and clockwise as negative.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000134 · EV0000135 · EV0000136 · +1 source units in pack.json

Core explanation · 15:00–19:59

Then you can take this for the horizontal component of the force and distance from here to here.

Key details

- You can also extend this line downwards till it touches the horizontal line at O.
- Then resolve this force along vertical.
- Determine what is the distance from here to here.
- Likewise, you can solve or find out the moment of this force.
- About the point O.
- See first method that I had talked about is this.
- You have 600 newtons, find out this D.
- That's one way of solving it.
- The other way of solving it is indicated here.
- You can resolve the force along X and Y as F1 and F2.
- You already know the distance 2 meters and 4 meters using this the 600 sine 40 into 2.
- Would be clockwise negative.
- This 600 cost 40 will also be clockwise into those four.
- Combination of these two would give you the moment about O.
- That's one way another way of solving it.
- Now next one is you extend this till it touches this axis.
- Then you resolve the force as we have done it here into F1 and F2.
- You will get F1 into this distance, if I can find this distance, okay, which is indicated as D1, then you can get the moment which will be clockwise.
- Next one is you extend this till it touches this horizontal line.
- Find out what is D2, then your F2 into D2, you will get the moment about O.
- This is one more way of solving it.
- This is another way of solving it.
- You can also try and solve it in one more way.
- Whatever you feel it is correct.
- I am not going to talk about it right now, but if you have any doubts, you can always meet me and then I will explain about that.
- The problem number is 2 bar 32.
- The problem statement is this.
- The force of magnitude F acts along the edge of the triangular plate.
- Determine the movement of F about O.
- And there is a force F that is acting, this is the force that is acting.
- What is asked is to determine the movement of F about O, that is this point.
- We can solve it in different ways.
- I can determine what is this angle.
- Similar the that angle will be same as this because this is alternate angles.
- Cos of this angle theta.

- Or I can also take this as a point of application.
- I can resolve this along this axis, which would be a side of this angle theta.
- Then that would into b will also give you give me the.
- Moment about O, or I can simply find out what is this perpendicular distance.
- In whichever way that you want, you can solve it.
- The other way of solving it is I can find out what is this.
- hypotenuse length is in terms of H and B, which is root of H square plus B square, that is what is done here.
- Next we will take up one more problem 2 bar 41.
- The problem statement is A 150 Newton pull T is applied to a card which is wound securely around the inner half of the drum.
- Determine the moment of T about the drum center C.
- Next is at what angle theta should T be applied so that the moment about the contact point is 0.
- The moment about this point should be 0.

Example or application from the lecture

- Now let us take another example from the unsolved problem.
- If I consider the point of application of this force as A.

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EV0000138 · EV0000139 · EV0000140 · +1 source units in pack.json

Core explanation · 20:00–24:59

What is that we have to take from this?

Key details

- Then from the diagram I can make out that the inner circle diameter or radius is 125.
- The outer circle radius is 200 millimeters.
- What are we supposed to find?
- First thing is movement about 0.0 because of this.
- Then second thing is at what angle theta we should apply this for such that moment at P is 0.
- What is the moment about.
- If I wound a wire around the circle and pull it, at whatever point that I want to pull, it will always be tangential to the circle.
- This you have to remember.
- It will always be tangential to the circle.
- Force that is P into the inner radius R1, and that's what is written here.
- Next, to find out at what angle theta this should act such that movement about this point is 0.
- If the force simply passes through.
- Or if I if I draw a line such that the line passes through B and also it is tangent to the inner circle.

- Why we should have that?
- Because there is no perpendicular distance.
- If the force passes through any point, the movement produced at that point is zero, that is because there is no perpendicular distance.
- The moment produced at C point.
- And that's what is done here.
- Moment at C is this, that should be equal to what?
- Your P cos theta into R.
- If I equate I'll get this.
- That is one way of solving it.
- The other way of solving it is if I take join this line, this line, this and the tangent.
- This, this, and this, this angle will be at night registration.
- I know this distance is 125, this distance is 200.
- I hope you have understood.
- We will stop today at this.
- We'll continue in our next class.
- That is in class number 9.
- My name is Dr.
- Sventesh from PS University.

Example or application from the lecture

- First thing is what is that we have to take from this given as the magnitude of the force P is 150 newtons.
- So let's take first one.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000142 · EV0000143 · EV0000144 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
UE26ME142A_Unit 1_Class 5_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/005_UE26ME142A_Unit_1_Class_5_AVSummary__86fffcdbe06dbb6b.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 5_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/005_UE26ME142A_Unit_1_Class_5_AVSummary__86fffcdbe06dbb6b.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 5_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/005_UE26ME142A_Unit_1_Class_5_AVSummary__86fffcdbe06dbb6b.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9996; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 5_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/005_UE26ME142A_Unit_1_Class_5_AVSummary__86fffcdbe06dbb6b.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 rendered pages in 001_UE26ME142A_Unit_1_Class_1.0_Lecture_PPT__a53eb8b08a5ac7ff.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 001_UE26ME142A_Unit_1_Class_1.1_Lecture_PPT__a6f76a088910b54a.slides.md have no readable `.slides.md` text; their original page images are included.
3. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.0_Lecture_PPT__de74ea957f21118c.slides.md have no readable `.slides.md` text; their original page images are included.

4. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.slides.md have no readable `.slides.md` text; their original page images are included.
5. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.slides.md have no readable `.slides.md` text; their original page images are included.
6. 3 rendered pages in 003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.slides.md have no readable `.slides.md` text; their original page images are included.
7. 3 rendered pages in 004_UE26ME142A_Unit_1_Class_4_Lecture_PPT__984b7c368532af06.slides.md have no readable `.slides.md` text; their original page images are included.
8. 3 rendered pages in 005_UE26ME142A_Unit_1_Class_5_Lecture_PPT__f49d361d79853022.slides.md have no readable `.slides.md` text; their original page images are included.
9. 2 rendered pages in 006_UE26ME142A_Unit_1_Class_6_Lecture_PPT__f79618b4a0f260c2.slides.md have no readable `.slides.md` text; their original page images are included.
10. 3 rendered pages in 008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.slides.md have no readable `.slides.md` text; their original page images are included.
11. 2 rendered pages in 009_UE26ME142A_Unit_1_Class_9_Lecture_PPT__832f697eaf1d383c.slides.md have no readable `.slides.md` text; their original page images are included.
12. 2 rendered pages in 010_UE26ME142A_Unit_1_Class_10.0_Lecture_PPT__aa8f331eb00d810f.slides.md have no readable `.slides.md` text; their original page images are included.
13. 2 rendered pages in 010_UE26ME142A_Unit_1_Class_10.1_Lecture_PPT__fe73ddc3ca596069.slides.md have no readable `.slides.md` text; their original page images are included.
14. 2 rendered pages in 011_UE26ME142A_Unit_1_Class_11_Lecture_PPT__3624142116c7273d.slides.md have no readable `.slides.md` text; their original page images are included.
15. 2 rendered pages in 012_UE26ME142A_Unit_1_Class_12.0_Lecture_PPT__2e7b61b526d51f60.slides.md have no readable `.slides.md` text; their original page images are included.
16. 4 rendered pages in 012_UE26ME142A_Unit_1_Class_12.1_Lecture_PPT__1d60317fe4af0cbe.slides.md have no readable `.slides.md` text; their original page images are included.
17. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.

18. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
19. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
20. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
21. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00030	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0056. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00031	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0056. Sidecar variants compare...
SRC00032	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0056. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00033	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0056. Sidecar variants compare...

C Revision and retrieval practice